Group Coursework Submission Form

Specialist Masters Programme

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Emerging Global Risks

Agent-Based Modelling of an Emerging Risk Scenario: A Systemic Analysis of Online Gambling Addiction Using NetLogo

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MSc Actuarial Science

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1 Introduction

This study builds the conceptual understanding of online gambling by employing an agent-based modelling (ABM) framework to simulate the progression and spread of gambling behaviour within a socially connected population. Rather than generalise trends or rely on aggregate assumptions, the model focuses on individual agents who make daily decisions influenced by both personal predisposition and peer behaviour. This approach helps us to see patterns like addiction, quitting and peer influence develop and change over time.

Using NetLogo as the simulation platform, the model introduces a set of configurable parameters such as base gambling addiction, addiction likelihood, and peer influence. These parameters help us examine how these variables interact to shape population-wide outcomes. Four distinct scenarios were tested, each representing different combinations of vulnerability and social contagion. These experimental conditions are grounded in evidence from the relevant literature and are designed to reflect real-world behavioural variability.

The purpose of this work is to evaluate how changes in individual and social parameters affect the incidence and trajectory of gambling addiction in a closed system. By analysing the simulation results, the study aims to identify conditions under which gambling-related harms become amplified or mitigated. The findings offer valuable insight for policymakers, researches, and public health stakeholders seeking to understand and manage risks associated with online gambling in digitally connected societies.

2 Background

Online gambling has become a major public health issue around the world. Not too long ago, it was something people only did at casinos or betting shops, but now it is a 24/7 activity that anyone can access from their phone. With the help of data and technology, online gambling platforms have grown fast and now reach hundreds of millions of people around the world. According to a 2024 report by Grand View Research, "*the global online gambling market size was valued at USD 63.53 billion in 2022 and is expected to grow at a compound annual growth rate (CAGR) of 11.7% from 2023 to 2030*" [1]. This rapid expansion highlights the speed with which the industry is growing, driven mainly by increased digital connectivity and unregulated platforms. As a result, several serious concerns have emerged about the broader impact of this trend on society.

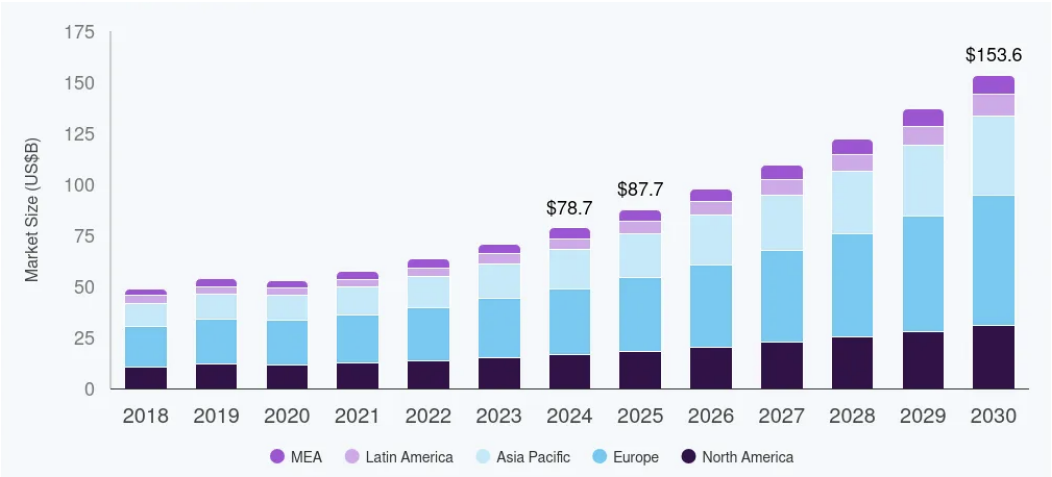

Figure 1: Online gambling market size by region, 2018–2030.[1]

The following are some of the key issues:

Financial Ruin: The pursuit of instant wealth through online gambling leads many individuals to chase losses, often spending more than they are capable. *"75% of gamblers and 53% of affected others who contacted the National Gambling Helpline reported financial difficulties, and 66% of gamblers said they were in debt"* (GamCare, 2021 [2]). As savings deplete and debt accumulates, some resort to selling possessions, highlighting the severe financial risks associated with gambling addiction.

Economic Impact: Online gambling diverts consumer spending away from productive sectors, negatively impacting the broader economy. A report by NERA Economic Consulting states, "*Between 2015-16 and 2022-23, British customers spent an average of £5.6bn annually on online gambling. We estimate that had the effect of reducing economic activity by £1.3bn per year and wages by £2.6bn.*"

Mental Health: Online gambling is closely linked to serious mental health issues such as depression, anxiety, addiction, and even suicidal thoughts. According to the Indian Journal of Psychiatry (2024)[3], "more than 90% of those with pathological

gambling have a psychiatric disorder, and more than 60% have three or more comorbid psychiatric disorders."

Cryptocurrency exploitation: The use of crypto largely increases addiction risk, by removing traditional bank safeguarding, enabling quick and anonymous bets. This is evident as "Crypto casinos have seen particularly rapid growth, with their market size increasing from $37 billion to over $70 billion by the end of 2024, and projections suggesting a rise to $150 billion by 2030." (DigitalFuel, 2024)[4].

3 Agent-Based Modelling Approach

3.1 Why Agent-Based Modelling?

Agent-Based Modelling is a computational approach for simulating autonomous and interacting agents to understand the behaviour of complex systems. The agents are programmed to behave and interact with other agents and the environment in specific ways. In the context of online gambling behaviour, ABM is suitable due to the heterogeneous nature of individuals, the non-linear social contagion effects, and the potential for emergent phenomena such as addiction clusters or behavioural tipping points.

ABM allows each gambler(agent) to be modelled autonomous decision-maker with unique characteristics such as their likelihood to gamble, susceptibility to addiction, and responsiveness to peer behaviour. These agents interact over a network, enabling researchers to simulate how gambling behaviours spread through social ties, how addiction develops over time, and how disengagement or relapse occurs.

Unlike statistical models or system dynamics, ABM supports non-linear dynamics, feedback loops, and emergent patterns such as addition clusters or sudden surges in gambling participation. This bottom-up approach makes it possible to investigate "what if" scenarios, assess risk factors, and test interventions (e.g.,reducing social influence, limiting exposure) in a controlled virtual environment.

Furthermore, ABM provides the behavioural granularity and flexibility required to realistically simulate the evolving dynamics of online gambling and addiction, making it a powerful tool for research, public health analysis, and policy evaluation.

3.2 NetLogo Platform Overview

NetLogo is a modelling platform that is especially useful for simulating complex human behaviours in connected systems. It was chosen for this study because it allows researchers to model individual gamblers as agents, each with their own traits and behaviours, such as the chance of gambling, becoming addicted, or stopping altogether. These agents can also be placed in a social network, making it possible to study how gambling and addiction spreads through peer influence.

NetLogo is easy to use and includes built-in tools for visualising results and running simulations under different conditions. This makes it ideal for exploring how gambling behaviour changes over time and in response to different factors, such as social pressure or individual vulnerability. The platform's flexibility and visual outputs make it a strong choice for studying the dynamics of online gambling and addiction in a realistic and controlled environment.

4 Model Design

This agent-based model simulates online gambling behaviour and the potential development of addiction within a socially connected population. A subset of agents starts as active gamblers, allowing gambling behaviour to spread from the very beginning of the simulation.

Each agent makes a daily decision on whether to gamble, based on their individual predisposition. Over time, agents may become addicted or lose interest and disengage. The likelihood of developing an addiction is influenced by social pressure, which depends on the number of gambling friends the agent has at any given moment.

The agents are linked through a random social network in which each agent is connected to at least three friends. The simulation runs for 365 time steps (days), allowing for the observation of the evolving dynamics of active gamblers, addicts, and disengaged individuals over time.

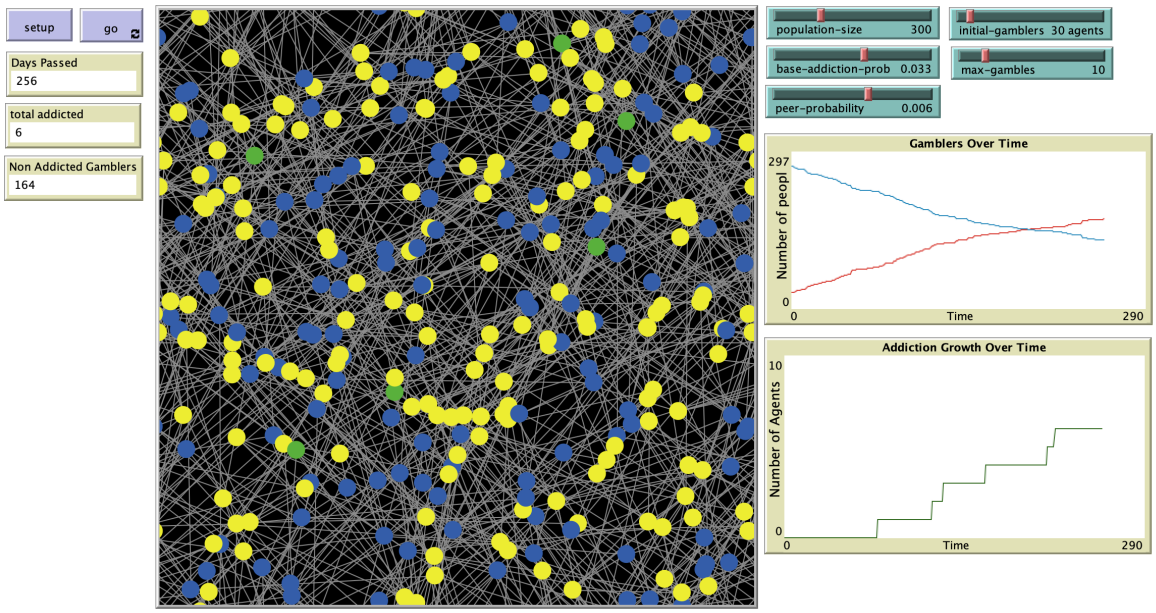

Figure 2: Model overview in Netlogo

4.1 Agents and Their Properties

Total Addicted:

Tracks the cumulative number of people who have developed a gambling addiction. It serves as a key metric to assess the impact of the spread of addictive behaviour within the social network. Its growth reflects both individual predisposition and social pressure.

Total Gamblers:

Indicates how many agents are actively gambling at any given time. This is useful for monitoring the daily dynamics of gambling behaviour, regardless of whether these agents eventually develop an addiction.

Gambling Count:

Tracks the number of times an agent has gambled. This variable allows for the evaluation of gambling exposure and can activate mechanisms like disinterest if the maximum allowed value is reached without addiction. It may also be interpreted as an indicator of cumulative risk.

Gamble Probability (Gamble Prob):

This is the individual probability that each agent will gamble in the absence of social influence. It reflects personal predisposition and varies among agents, introducing behavioural diversity. Combined with social pressure, it determines the final decision to gamble.

4.2 Environment and Interactions

Agents are linked through a random social network where each agent is connected to at least three friends. Social influence is incorporated through a simple mechanism: every time an agent has friends who are gambling, their probability of doing so increases, depending on how many of those friends are active.

An agent's behaviour, therefore, is not solely determined by personal predisposition but also by their surrounding environment, allowing the model to simulate social contagion. As Forsström et al. (2022)[5] note:

"Our results not only reaffirm that gambling participation and support among peer groups and family is connected to increased gambling harm."

To simplify this dynamic, each gambling friend increases an agent's addiction probability by 0.003 (equivalent to an additional 0.3%). This value represents one-third of the total effect observed in studies showing that addiction risk doubles when the entire social environment is gambling.

4.3 Key Parameters and Assumptions

Initial Gamblers:

Defines how many agents start the simulation as active gamblers. These serve as "seeds" that trigger the spread of gambling behaviour throughout the network. Since an agent's gambling probability increases with the presence of gambling friends, initial gamblers are crucial for diffusion from the start.

Base Gamble Probability:

To define the daily gambling dynamic, it is assumed that the average individual in the U.S. has an 11% chance of gambling at least once per year, based on official data from the Pennsylvania Department of Drug and Alcohol Programs (2024)[6].

Given that each tick represents one day, this annual probability is converted into a daily probability:

$$P_{\text{daily}} \approx 1 - (1 - 0.11)^{1/365} \approx 0.0003\,(0.03\%)$$

This value is used as the base gamble probability in the simulation.

Base Addiction Probability:

Represents the baseline probability that an agent becomes addicted after gambling. A higher value implies a more vulnerable population and accelerates the emergence of addicted individuals. A range of 1.5% to 5% will be used, based on findings from the PubMed Central (2015) study[7], which reported this prevalence range among youth and adults in Asia.

Max Gambles:

Sets the maximum number of times an agent can gamble without becoming addicted. If this threshold is reached without addiction, the agent is assumed to have lost interest. Based on Okamura et al. (2015)[8], approximately 15.5% of individuals who relapse engage in 10 or more gambling episodes. Therefore, the threshold is set at 10. Reaching this value without stopping indicates a transition to addiction, while not reaching it may represent disinterest or impulse control.

5 Scenario-Based Simulation Results and Observations

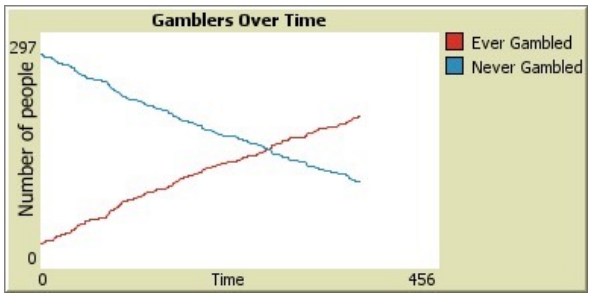

Figure 3: Cumulative Transition from Non-Gamblers to Gamblers Over Time

This section outlines the results generated from the agent-based simulation constructed in NetLogo. The core objective is to assess how variations in base addiction probability and peer influence affect the development of addictive behaviours over time.

To ensure consistent comparisons across different conditions, several key parameters were held constant throughout all simulations, as shown in the table below.

Parameter	**Value**
Population Size	300
Initial Gamblers	30
Max Gambles	10
Probability Gambles	0.3%
Simulation Duration	365 days (ticks)

Table 1: Key Simulation Parameters

We tested four distinct scenarios by changing two independent variables:

- **Base Addiction Probability:** Likelihood that an individual becomes addicted each time they gamble.
- **Peer Influence Probability:** Additional probability of becoming addicted due to exposure from peers who gamble.

These values were selected based on empirical evidence from international studies on gambling addiction prevalence. In particular, a study published in PubMed Central (2015) [7] reported that problem gambling rates among youth and adults in various Asian countries ranged from 1.5% to 5%, informing our selection for the base addiction probabilities. For peer influence, we used values of 0.3% and 0.6%, representing low and moderate levels of social reinforcement, respectively.

Based on these assumptions, the following four experimental scenarios were tested:

Scenario	Base Addiction Probability	Peer Influence Probability
1	1.5%	0.3%
2	5%	0.3%
3	1.5%	0.6%
4	5%	0.6%

Table 2: Experimental Scenarios Tested in the Simulation

After running each of the four scenarios three times and averaging the results, the following trends were observed:

Figure 4: Scenario 1: Base Addiction Probability: 1.5%, Peer Influence: 0.3%

Figure 5: Scenario 2: Base Addiction Probability: 5%, Peer Influence: 0.3%

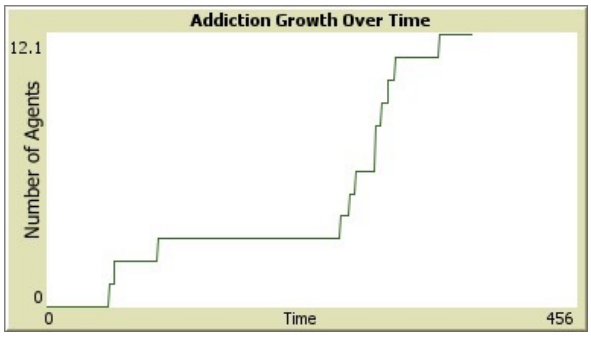

Figure 6: Scenario 3: Base Addiction Probability: 1.5%, Peer Influence: 0.6%

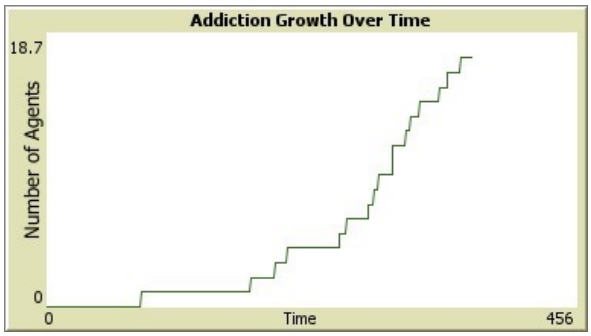

Figure 7: Scenario 4: Base Addiction Probability: 5%, Peer Influence: 0.6%

Scenario	Base Addiction Prob	Peer Influence Prob	% Gambled	% Addicted
1	1.5%	0.3%	66.4%	2.8%
2	5%	0.3%	65.7%	5.3%
3	1.5%	0.6%	69.9%	3.8%
4	5%	0.6%	63.7%	6.6%

Table 3: Summary of Experimental Scenarios and Results

The simulation results show that gambling activity spreads rapidly, with over 63% of the population engaging in gambling at least once across all four scenarios. This occurred despite a low initial gambling probability (0.3%), suggesting that the ease of online access plays a powerful role in how quickly gambling can spread. Looking at the graph, we can see a rapid early growth in people who have gambled that gradually slowed as the gambling population approached saturation.

Addiction development followed a very different pattern, where addiction rates began slowly because only a small group had started gambling initially. Then, as gambling spread, addiction levels rose significantly. This acceleration happened because more gamblers meant more people were at risk, while exposure over time increased vulnerability. The delay between the rise in gambling and the surge in addiction suggests this can be stopped with early intervention.

Comparing scenarios with the same peer influence (0.3%) but different base addiction probabilities shows how much individual vulnerability matters. For example, Scenario 1 (1.5% base addiction probability) resulted in 2.8% addiction level, while Scenario 2 (5.0% base addiction probability) led to 5.3% addiction level. This 135% increase demonstrates the strong impact of individual risk factors and also raises concerns about modern digital gambling platforms, which may raise this risk further through design features that encourage longer engagement.

When base addiction probability was held constant, peer influence acted as a booster rather than a primary cause. It did not significantly increase overall gambling participation but did raise addiction rates among existing gamblers. For example, when peer influence doubled from 0.3% to 0.6% in low-risk scenarios (1.5% base probability), addiction rose from 2.8% to 3.8%, while in high-risk scenarios (5.0% base

probability), addiction increased from 5.3% to 6.6%. This suggests that peer influence doesn't expand the overall gambling market as much as it pushes more people from casual gambling into addiction.

6 Policy Recommendations and Strategic Insights

The results of this simulation have important consequences for how governments and regulators should manage the growing risks of online gambling addiction. The findings show that even when the chance of addiction is low, and social influence is moderate, a large percentage of the population can still be affected. This suggests that online gambling is not just a personal issue, but a public health concern that can spread quickly in society, especially in digitally connected communities.

One major insight is that easy access to online gambling is a key reason why gambling spreads so fast. In the simulation, more than 63% of the population gambled at least once, even though the starting probability of gambling was only 0.3%. This reflects what is happening in the real world i.e. people can now gamble at any time from their phones or computers, with very few restrictions. To reduce this harm, regulators should consider rules that limit how much and how often people can gamble online. These rules could include daily time limits, deposit caps, session breaks, and tools to lock people out after harmful patterns are detected—especially for younger users.

Secondly, the model showed that peer influence (being around other gamblers) plays an important role in increasing addiction. Although peer pressure didn't cause many more people to start gambling, it did increase the number of people who became addicted. For example, when peer influence was doubled, addiction levels rose by as much as 25% in some scenarios. This means that peer effects do not create new gamblers but push casual ones toward addiction. Therefore, policies should aim to reduce the social spread of gambling, for instance, by banning referral bonuses, influencer gambling ads, and platform features that encourage users to invite friends.

Another important point is that individual risk matters. When we compare simulations where peer influence is the same but addiction risk is different, the results clearly show that more vulnerable individuals like those with higher base addiction probability end up with much higher addiction levels. This supports real-world research showing that young men, low-income groups, and people with mental health challenges are more likely to suffer from gambling addiction. Policymakers should create targeted support programs for these groups, such as early education in schools, mental health screening, and free or low-cost counselling services for high-risk users.

The model also highlights the need for better use of data and technology. Governments should require gambling companies to share anonymous user data with researchers and public health institutions. This would help in building better models, tracking risks, and improving regulations. In addition, companies should be forced to build systems that monitor risky behaviour in real-time such as long gambling sessions, large financial losses, or gambling late at night—and automatically take action, like

suggesting a break or limiting access.

Finally, regulators must adopt a more flexible and responsive approach. Gambling addiction does not grow in a straight line. It often stays hidden and then grows quickly after certain tipping points, like exposure to gambling ads or peer pressure. Policies should be able to respond quickly when problems begin to grow. This could include pilot programs, live testing of new rules, and better cooperation between health agencies, educators, and technology experts.

In short, this simulation shows that tackling online gambling addiction requires a broad and well-coordinated effort. Simple rules or awareness campaigns are not enough. Instead, policymakers need a mix of technology, targeted support, regulation, and real-time monitoring to reduce the spread and impact of addiction in the digital age.

7 Model Limitations

While the agent-based model provides valuable insight into the diffusion and escalation of online gambling behaviour, several structural and theoretical limitations constrain its applicability.

7.1 Simplified Social Network Structure

The model initialises a social network where each agent is randomly connected to at least three peers. While this creates a basic structure for simulating peer influence, it simplifies real-world social networks, which are often scale-free, clustered, or structured hierarchically. In actual gambling environments, some individuals (like social media influencers or highly connected users may) exert a disproportionate impact on behavioural spread. The current model does not allow for such influence over the social graph over time. As a result, it may under-represent the potential for rapid contagion through tightly connected communities or central actors, thus limiting the model's ability to replicate realistic diffusion dynamics seen in online gambling behaviour.

7.2 Absence of Demographic-Specific Vulnerability

The current model treats agents homogeneously apart from variations in base gambling probability, omitting important demographic risk factors such as age, gender, and socio-economic status. However, global research consistently shows that certain groups (especially young adult males) are significantly more prone to online gambling addiction. For instance, Calado and Griffiths (2016)[9] found that younger age and male gender were the most consistent demographic predictors of problem gambling across 69 global studies. Similarly, Dellosa and Browne (2024)[10] reported that men were over three times more likely than women to develop gambling problems, and that individuals under 35 faced elevated risk. By not modelling demographic-specific vulnerabilities, the simulation risks overlooking the concentrated nature of harm among key subpopulations, which is critical for designing effective policy responses.

7.3 No Financial or Emotional Variables Tracked

A major limitation of the model is its omission of financial stress and psychological distress, which are both key dimensions of real-world gambling harm. In the simulation, agent decisions are based solely on gambling frequency, addiction status, and social influence. However, some research, such as GamCare (2022)[2] and Raj, Sharma, and Kaur (2024)[3], consistently shows that online gambling addiction is deeply intertwined with monetary loss, debt accumulation, anxiety, depression, and in some cases, suicidal ideation.

By not modelling any economic or emotional metrics the current simulation cannot capture how financial desperation, or emotional crises may accelerate addiction, trigger relapse, or hinder recovery. Likewise, it lacks mechanisms to simulate how interventions (e.g. debt support or mental health counselling) could reduce harm or aid in disengagement.

7.4 Static and Binary Addiction States

Addiction in the model is implemented as a binary switch: agents are either addicted or not, with no intermediate or fluctuating states. This rigid behavioural structure contrasts with findings from clinical and longitudinal research showing that gambling addiction is cyclical and dynamic. Grall-Bronnec et al. (2025)[11] observed that addiction often progresses through repeated phases of control, relapse, and partial recovery. The omission of this temporal fluidity results in an oversimplified model that fails to capture the lived reality of many with gambling disorders. A more nuanced state framework (e.g., incorporating relapse propensity, recovery trajectories, or intensity levels) would enhance realism and policy relevance.

While the agent-based model offers a useful framework for simulating the spread of online gambling through social influence and behavioural contagion, it necessarily involves simplifications that constrain its realism. These limitations do not invalidate the model's insights but highlight areas where future iterations could integrate richer data, behavioural nuance, and dynamic adaptation to better reflect the lived experiences of individuals affected by online gambling addiction. Enhancing these aspects would improve the model's utility for policy design and targeted intervention strategies.

8 Conclusion

The aim of this report was to examine the emerging global risk of online gambling addiction using agent-based modelling (ABM) in NetLogo. As gambling platforms become more accessible and socially embedded, the potential for widespread addiction grows. Through this study, we sought to simulate how individual susceptibility and social influence interact to drive gambling behaviour in digitally connected environments.

Our objectives have been achieved by:

- Simulating gambling and addiction dynamics over time
- Assessing the influence of peer influence and individual vulnerability
- Testing multiple scenarios to evaluate regulatory effectiveness
- Outlining practical and evidence-based intervention strategies

Simulation results demonstrated that even modest peer influence and addiction probabilities can result in widespread gambling participation and steadily rising addiction levels. Critically, the time lag between gambling uptake and addiction onset suggests a window for early intervention before harm escalates.

The most effective scenario combined lower individual susceptibility with reduced social reinforcement, supporting the view that coordinated intervention strategies are more impactful than isolated measures. We recommend that future models incorporate demographic and emotional complexity to more accurately reflect real-world dynamics.

While ABM offers powerful insights, we acknowledge that behavioural simplifications and ethical blind spots limit its scope. Effective management of online gambling risk will require proactive regulation, platform accountability, and sustained cross-sector collaboration to protect at-risk populations in an increasingly digital society.

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A Appendix

A.1 Agent-Based Gambling Simulation

```

undirected-link-breed [friendships friendship]

globals [
  total-addicted
  total-gamblers
]

turtles-own [
  addicted?          ; true if the agent is addicted
  gambling?          ; true if the agent is gambling this tick
  gamble-prob        ; base probability of gambling
  gambling-count     ; number of times the agent has gambled
  lost-interest?     ; true if the agent lost interest
  ever-gambled?      ; true if the agent has ever gambled
]

to setup
  clear-all
  set-default-shape turtles "circle"

  create-turtles population-size [
    setxy random-xcor random-ycor
    set addicted? false
    set gambling? false
    set gamble-prob 0.003
    set gambling-count 0
    set lost-interest? false
    set ever-gambled? false
    set color blue
  ]

  ; Randomly connect each agent with 3 distinct friends
```

```
  ask turtles [
    create-friendships-with n-of 3 other turtles
  ]

  ; Initialize the initial gamblers (seeds of the behavior)
  let seed-count min (list initial-gamblers population-size)
  ask n-of seed-count turtles [
    set gambling? true
    set ever-gambled? true
    set color yellow
  ]

  set total-addicted 0
  reset-ticks
end

to go
  if ticks >= 365 [ stop ]

  ask turtles [
    set gambling? false
    if not addicted? and not lost-interest? and ever-gambled? [ set color yellow ]
    if not addicted? and not lost-interest? and not ever-gambled? [ set color blue ]
  ]

  set total-addicted count turtles with [addicted?]
  set total-gamblers 0

  ask turtles [
    if not addicted? and not lost-interest? [
      let total-prob gamble-prob
      if random-float 1 < total-prob [
        gamble
      ]
    ]
  ]

  set total-gamblers count turtles with [color = yellow]
  tick
end

to gamble
  set gambling? true
  set ever-gambled? true
  set gambling-count gambling-count + 1

  ; Social influence only affects the probability of becoming addicted
```

```
  let friends-gambling count link-neighbors with [ever-gambled ?]
  let peer-addiction-influence friends-gambling * peer-probability  ; Adjustable coefficient
  let total-addiction-prob min (list 1 (base-addiction-prob + peer-addiction-influence))

  if not addicted? and (random-float 1 < total-addiction-prob) [
    set addicted? true
    set color green
  ]

  if gambling-count >= max-gambles and not addicted? [
    set lost-interest? true
    set gambling? false
    set color gray
  ]
end
\begin{figure}
    \centering
    \includegraphics[width=0.5\linewidth]{Modeloverview.png}
    \caption{Enter Caption}
    \label{fig:enter-label}
\end{figure}
```

Listing 1: NetLogo code for the agent-based gambling behaviour model